SHUBHRAM PRIYADARSHAN

QA Automation Engineer | Data Science (AI/ML) Student

+49 1791554579 | shubhrampriyadarshan1@gmail.com | Germany, Nuremberg
github.com/shbzstrkr1999 | linkedin.com/in/shubhram-priyadarshan | gitlab.com/shbzstrkr1

SUMMARY

QA Automation Engineer with almost 5 years of professional experience in software quality assurance and test automation, including engineering a Python/Pytest framework to validate AI judge agents assessing downstream model performance. Currently pursuing a Master's in Data Science (AI/ML), and building hands-on software development and machine learning experience through independent projects: a full-stack AI learning platform backend using FastAPI and the Gemini API, and a machine learning classification pipeline using Scikit-learn and Random Forest. Strong interest in AI-driven infrastructure, backend systems, and reliability validation of AI-based systems. Expertise in Python, Java, SQL, FastAPI, JUnit, Jira, Scikit-learn, and API design.

SKILLS

Development & Languages: Python, Java, SQL, FastAPI, REST APIs, SQLAlchemy

Data Science & Machine Learning: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, GridSearchCV, AI Model Validation, Gemini API Integration

DevOps & Databases: GitLab CI/CD, Docker, SonarQube, MongoDB, MySQL

Project Management & Documentation: Jira, Confluence

Testing & Tools: Selenium, TestNG, Pytest, JUnit, API Testing, Postman, Performance Testing, JMeter, Security Testing, Regression Testing, TestRail, BackstopJS

EXPERIENCE

Software Automation Engineer II – IQVIA-IMS Health Analytics, India
03/2023 – 03/2026

- **Engineered** a Python and Pytest automation framework to validate AI "judge agents" evaluating downstream model performance, seamlessly integrating results into TestRail.
- **Built and maintained** 200+ automated test cases with Selenium WebDriver, Java, and TestNG for enterprise healthcare applications, expanding regression coverage from 50 to 200+ tests and reaching 75% automation coverage.
- **Cut** regression testing time by 50% from 4 days to 2 days through automation improvements, helping accelerate release cycles and improve QA efficiency.
- **Owned** the automation strategy as the sole QA Automation Engineer on a 3-person manual QA team, guiding framework enhancements and coverage expansion.
- **Designed** reusable utility classes and improved the automation framework to strengthen maintainability, scalability, and execution reliability.
- **Executed** automated suites in headless GitLab CI/CD pipelines, configuring YAML files and runners to support multiple test environments.
- **Performed** functional, regression, integration, end-to-end, API, and visual regression testing with Postman and BackstopJS to validate stability, data integrity, and cross-browser compatibility.
- **Tracked** defects with Agile cross-functional teams and used SonarQube for code quality analysis and continuous QA process improvement.

Software Test Engineer – Saveo Healthtech Pvt Ltd, India
04/2021 – 02/2023

- **Performed** functional, regression, and exploratory testing of Android applications across multiple devices, ensuring application stability, usability, and cross-device compatibility.
- **Executed** SQL-based database validation and backend testing to verify data integrity, troubleshoot issues, and ensure accurate system behavior.
- **Developed** and maintained automation scripts for sanity and regression testing, improving test efficiency and supporting reliable software releases.
- **Collaborated** with product owners, business analysts, and developers to analyze requirements, refine functional specifications, and contribute to a 25% improvement in operational efficiency.
- **Independently owned** end-to-end quality assurance for the Inventory Management System, validating inventory workflows, returns processing, damaged-item handling, and real-time inventory updates through scheduled cron jobs.
- **Conducted** performance and load testing for a customer gamification platform, ensuring production readiness and supporting a successful launch that contributed to a 30% increase in GMV.

- **Validated** microservices-based applications through API testing, JWT authentication verification, backend workflow testing, and inter-service communication checks, while receiving two "Saviour of the Month" awards for outstanding contributions and product improvement initiatives.

EDUCATION

SRH University, Germany — Present
Post Graduation, Master's in Data Science (AI/ML)

Presidency University, India — 2021
Bachelor of Technology, Computer Science

PROJECTS

LectAI – AI Powered Learning Platform (Live Demo: edu-ai-platform-n34i.onrender.com)
Python, FastAPI, React, TypeScript, MySQL, SQLAlchemy, Docker, Gemini API

- **Built and deployed** a full-stack AI learning platform enabling students to upload lecture materials and generate AI-powered summaries and quizzes.
- **Developed** a modular FastAPI backend using layered architecture with REST APIs, SQLAlchemy ORM, JWT authentication, and MySQL.
- **Built** a React + TypeScript frontend supporting session authentication, lecture management, and AI-generated study resources.
- **Integrated** Google's Gemini API for lecture summarization and quiz generation with structured response handling.
- **Containerized** the application using Docker and deployed frontend and backend independently while maintaining a unified repository structure.
- **Designed** the architecture to cleanly support future scalability while operating stably as a modular monolith.

Water Potability Prediction – Machine Learning
Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Jupyter Notebook

- **Built** an end-to-end machine learning pipeline to classify 3,276 water samples as potable or non-potable using 9 physicochemical water quality features.
- **Applied** KNN Imputation to handle missing values, selecting it over mean/median imputation for better preservation of feature distributions.
- **Conducted** Exploratory Data Analysis and correlation analysis, identifying weak linear relationships and dataset imbalance (61% non-potable, 39% potable).
- **Trained and evaluated** Logistic Regression and Random Forest classifiers, improving ROC-AUC from 0.5482 to 0.6470 by capturing non-linear feature interactions.
- **Optimized** the Random Forest model using GridSearchCV with 5-fold cross-validation, achieving 65.85% accuracy and an F1 score of 0.4136 as the best-performing model.

LANGUAGES

English (C1) — German (A2)